황준오 Juno Hwang

Affiliation

Department of Physics Education, Data Science Lab, Seoul National University
Ph.D. Candidate (All but dissertation)

Contact

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Keywords Statistical physics, Data science, Physics education, Learning science, Diffusion models

Education

Seoul National University

, Seoul, Korea

Integrated M.S./Ph.D., Physics Education & Data Science Lab, 2021–present (All but dissertation)

B.S., Computational Science, February 2021

Face and gaze recognition based mouse control

B.S., Physics Education, February 2021

Approximation of boundary conditions for low-frequency resonance of low-dimensional wind instruments

Career

Professional Research Personnel (Alternative Military Service), 2024–present (Currently on deferral)

LBOX Co., Ltd.

Research Intern, December 2021–February 2022

Legal document image augmentation and classification

OCR correction algorithm development

Vittgen Inc.

Technical Director, 2017–2020

Overall development of games and interactive media

License & Certificate

Teacher's Certificate, Secondary School Regular Teacher (Grade II) of Physics, February 2021

Teaching Experience

Seoul National University, Department of Science Education, Instructor for *Science Education Forum* (2026)

Seoul National University, Department of Physics Education, Teaching Assistant for *Computational Physics and Education* (2023)

Seoul National University, Center for Liberal Education, Head Teaching Assistant for *Introduction to Computing* (2019–2023)

Papers

S. Hong, J. Moon, T. Eom, **J. Hwang** & J. Seo (2026). *Leveraging learning analytics to enhance immersive teacher simulations: Challenges and opportunities.* Preprint (arXiv).

S. Hong, J. Moon, T. Eom, I. D. Awoyemi & **J. Hwang** (2025). *Generative AI-Enhanced Virtual Reality Simulation for Pre-Service Teacher Education: A Mixed-Methods Analysis of Usability and Instructional Utility for Course Integration. Education Sciences* (MDPI).

Y. Xie, **J. Hwang**, A. Kim, H. Kim & Y. H. Cho (2025). *Effect of AI-Powered Virtual Reality Simulation on Pre-Service Teachers' Empathy towards Multicultural Students.* International Conference of the Learning Sciences (ICLS) 2025.

J. Hwang, S. Hong, T. Eom & C. Lim (2024). *Enhancing Pre-service Teachers' Competence with a Generative Artificial Intelligence-Enhanced Virtual Reality Simulation*. International Society of the Learning Sciences (ISLS) Annual Meeting 2024.

J. Hwang, Y. H. Park & J. Jo (2024). *Upsample guidance: Scale up diffusion models without training*. Preprint (arXiv).

J. Hwang, Y. H. Park & J. Jo (2023). *Resolution chromatography of diffusion models*. Preprint (arXiv).

H. Soh, D. Kim, **J. Hwang** & J. Jo (2023). *Mirror descent of Hopfield model. Neural Computation* (MIT Press), 35(9), 1529–1542.

J. Hwang, W. S. Hwang & J. Jo (2020). *Tractable loss function and color image generation of multinary restricted Boltzmann machine*. NeurIPS 2020 DiffCVGP Workshop.

Books & Chapters

S. Hong, J. Moon, T. Eom, **J. Hwang**, J. Lim & S. Park (2026). *Immersive and Engaging Design for Teacher Simulation: Theoretical Foundations and Innovative Approaches*. In P. Trifonas (Ed.), *International Handbook of Theory and Research in Digital Media and Education*. Springer.

Y.-T. Lee, K.-M. Kim, Y.-O. Choi, J.-W. Han, **J. Hwang**, Y.-G. Lee, Y.-N. Kim, Y.-J. Lee & I.-T. Kim (2022). *Nonbibeoseo: Schema Concept Dictionary for CSAT Korean — Science and Technology*. Hyongseol EMJ.

Invited Talks

2024 KIAS CAC Summer School on the Parallel Computing and Artificial Intelligence — *Seoul*

2023 KIAS CAC Summer School on the Parallel Computing and Artificial Intelligence — *Seoul*

2023 KIAS Center for AI and Natural Sciences Winter Workshop — *Gangwon*

2022 KIAS Winter School for Physics and Machine Learning — Generalization and extended models of Boltzmann machines — *Online*

2021 APCTP Workshop for Physics and Machine Learning — *Jeju*

2019 Hyundai Engineering In-house AI Research Group Invited Talk — *Seoul*

Conference Presentations

J. Hwang & J. Jo (2026). *Nonequilibrium Gated Simulated Annealing*. 2026 KPS Spring Meeting, Daejeon.

H. K. Park, **J. Hwang**, S. Hong & S. N. Martin (2024). *Facilitating High School Students' Scientific Modeling through Interaction with AI Chatbots with Varied Personas*. International Conference on Science Education in the Infosphere, Seoul.

J. Hwang, S. Hong, T. Eom & C. Lim (2024). *Enhancing Pre-service Teachers' Competence with a Generative Artificial Intelligence-Enhanced Virtual Reality Simulation*. ISLS Annual Meeting, Buffalo.

J. Hwang, D. Lee, S. Kwak, K. Kim, D. Jeong & J. Cho (2024). *Noise- and intensity-robust peak* *detection for machine-learning-based classification of organic dye SERS signals*. Korean Society for Heritage Conservation Spring Conference, Seoul.

J. Hwang & J. Jo (2024). *Mode Separating Property of Restricted Boltzmann Machines*. 2024 KPS Spring Meeting, Daejeon.

J. Hwang, S. Hong, T. Eom & C. Lim (2023). *Development of generative AI-based virtual reality classroom simulation for enhancing pre-service teachers' instructional competence*. Joint Fall Conference of KASET & KEMIE, Seoul.

J. Hwang & J. Cho (2022). *Design and application of spring motion lesson with real-time remote physics data collection*. Korean Society for Field Science Education Annual Conference, Siheung.

J. Yoo, J. Park, D. Lee, S. Jin & **J. Hwang** (2018). *Visualization of Real Magnetic Field Using Sensor and AR*. American Association of Physics Teachers Winter Meeting, California.

Research Projects

AI-based thermal degradation prediction technology for stone cultural heritage under climate change (2026.04–present) — National Research Institute of Cultural Heritage

Causal inference of time series (2022.12–2026.02) — Ministry of Science and ICT

In-situ analysis, diagnosis, and degradation prediction technology for organic dyes (2021.07–2026.02) — National Research Institute of Cultural Heritage

Development and application of webcam-based eye tracking for analyzing English learning processes in digital environments (2024.12–2025.12) — Seoul National University

Future education innovation and blended creative convergence education based on intelligent information technology (2021.06–2025.03) — Korea Foundation for the Advancement of Science and Creativity

Virtual reality simulation development for multicultural education competence (2023.12–2024.02) — Seoul National University

Revised course for common graduate curriculum — Introduction to machine learning for AI convergence education (2023.12–2024.01) — Seoul National University

Virtual reality classroom simulation for enhancing pre-service teachers' instructional competence (2023.06–2023.08) — Seoul National University

VR simulation environment for undergraduate physics education demonstration experiments (2022.11–2023.02) — Seoul National University

Dynamical inference of time series data (2020.01–2022.11) — National Research Foundation of Korea

Development

Juno-DKT (Deep Knowledge Tracing) pytorch implementation — PyTorch implementation of Deep Knowledge Tracing

SNUWET — Webcam-based screen recording and AI eye-tracking automation platform for web browsers

Awards

Best Paper Award (2023 Joint Fall Conference of KASET & KEMIE) — Development of generative AI-based virtual reality classroom simulation for enhancing pre-service teachers' instructional competence

Award (2018 Korea East-West Power Big Data & AI Contest) — Solar power generation prediction model by climate

Excellence Award (2016 KAIST Wearable Computer Contest) — Team Gorany: EMG and IMU sensor based electronic instrument LaunchWare

First Prize (2013 Ministry of Science, ICT and Future Planning Big Data Fair) — Daily movie audience prediction algorithm based on pre-release reactions

Silver Prize (2012 Korea Wi.Content Contest) — Game development category *Amazing Shiri*